

DPP NO.-6

- Q.1 Number of different ways in which 8 different books can be distributed among 3 students, if each student receives atleast 2 books is _____.
- Q.2 There are 10 seats in a double decker bus, 6 in the lower deck and 4 on the upper deck. Ten passengers board the bus, of them 3 refuse to go to the upper deck and 2 insist on going up. The number of ways in which the passengers can be accommodated is _____. (Assume all seats to be duly numbered)
- Q.3 Find the number of permutations of the word "AUROBIND" in which vowels appear in an alphabetical order.
- Q.4 The greatest possible number of points of intersection of 9 different straight lines & 9 different circles in a plane is:
(A) 117 (B) 153 (C) 270 (D) none
- Q.5 An old man while dialing a 7 digit telephone number remembers that the first four digits consists of one 1's, one 2's and two 3's. He also remembers that the fifth digit is either a 4 or 5 while has no memorising of the sixth digit, he remembers that the seventh digit is 9 minus the sixth digit. Maximum number of distinct trials he has to try to make sure that he dials the correct telephone number, is
(A) 360 (B) 240 (C) 216 (D) none
- Q.6 If as many more words as possible be formed out of the letters of the word "DOGMATIC" then the number of words in which the relative order of vowels and consonants remain unchanged is _____.
- Q.7 Number of ways in which 7 people can occupy six seats, 3 seats on each side in a first class railway compartment if two specified persons are to be always included and occupy adjacent seats on the same side, is $(5!) \cdot k$ then k has the value equal to :
(A) 2 (B) 4 (C) 8 (D) none
- Q.8 Number of ways in which 9 different toys be distributed among 4 children belonging to different age groups in such a way that distribution among the 3 elder children is even and the youngest one is to receive one toy more, is :
(A) $\frac{(5!)^2}{8}$ (B) $\frac{9!}{2}$ (C) $\frac{9!}{3!(2!)^3}$ (D) none
- Q.9 In an election three districts are to be canvassed by 2, 3 & 5 men respectively . If 10 men volunteer, the number of ways they can be allotted to the different districts is :
(A) $\frac{10!}{2! 3! 5!}$ (B) $\frac{10!}{2! 5!}$ (C) $\frac{10!}{(2!)^2 5!}$ (D) $\frac{10!}{(2!)^2 3! 5!}$
- Q.10 Let P_n denotes the number of ways in which three people can be selected out of 'n' people sitting in a row, if no two of them are consecutive. If, $P_{n+1} - P_n = 15$ then the value of 'n' is :
(A) 7 (B) 8 (C) 9 (D) 10

- Q.11 Number of ways in which 8 people can be arranged in a line if A and B must be next each other and C must be somewhere behind D, is equal to
 (A) 10080 (B) 5040 (C) 5050 (D) 10100
- Q.12 A has 3 maps and B has 9 maps. All the 12 maps being distinct. Determine the number of ways in which they can exchange their maps if each keeps his initial number of maps.
- Q.13 Number of three digit number with atleast one 3 and at least one 2 is
 (A) 58 (B) 56 (C) 54 (D) 52

Paragraph for Question Nos. 14 to 16

16 players $P_1, P_2, P_3, \dots, P_{16}$ take part in a tennis tournament. Lower suffix player is better than any higher suffix player. These players are to be divided into 4 groups each comprising of 4 players and the best from each group is selected for semifinals.

- Q.14 Number of ways in which 16 players can be divided into four equal groups, is
 (A) $\frac{35}{27} \prod_{r=1}^8 (2r-1)$ (B) $\frac{35}{24} \prod_{r=1}^8 (2r-1)$ (C) $\frac{35}{52} \prod_{r=1}^8 (2r-1)$ (D) $\frac{35}{6} \prod_{r=1}^8 (2r-1)$
- Q.15 Number of ways in which they can be divided into 4 equal groups if the players P_1, P_2, P_3 and P_4 are in different groups, is :
 (A) $\frac{(11)!}{36}$ (B) $\frac{(11)!}{72}$ (C) $\frac{(11)!}{108}$ (D) $\frac{(11)!}{216}$
- Q.16 Number of ways in which these 16 players can be divided into four equal groups, such that when the best player is selected from each group, P_6 is one among them, is $(k) \frac{12!}{(4!)^3}$. The value of k is :
 (A) 36 (B) 24 (C) 18 (D) 20